Ionospheric variations associated with stratospheric sudden warmings: current understanding and future challenges

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Seminar at Boston College, May 24, 2011

Outline

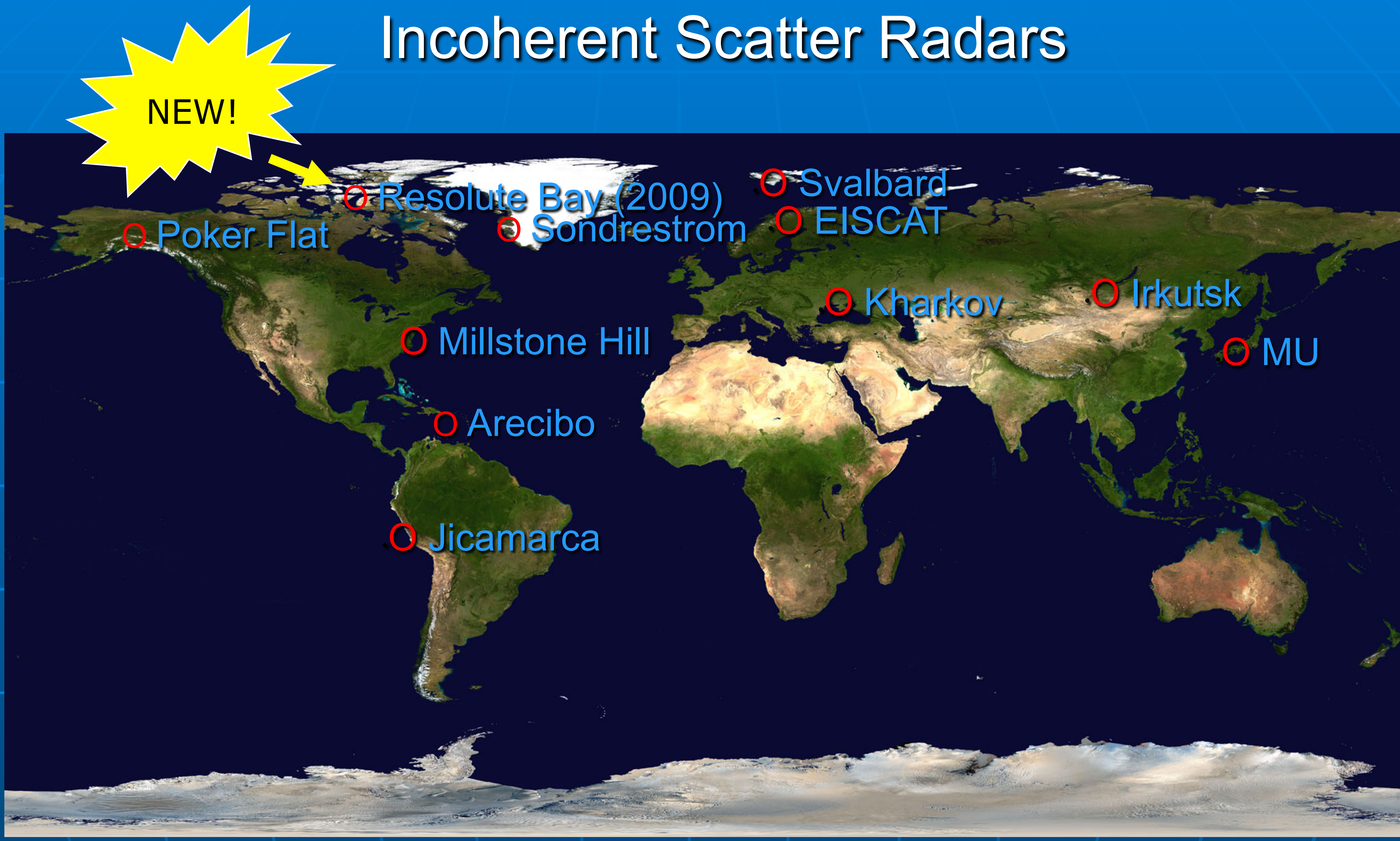
- Stratospheric sudden warming: what is it?
- Why are we interested in stratwarmings?
- Summary of ionospheric effects associated with stratospheric sudden warmings
  - Focus on the upper atmosphere (h > 100 km)
  - Focus on experimental results
- Interpretation of observations
- Questions arising from initial results
- Implications
- Outlook for the future studies

Millstone Hill Incoherent Scatter Radar

- Westford, MA, 42.6N, 288.5E
- Part of MIT Haystack Observatory
- Operational since 1960s
- Zenith antenna – 68 m
- Steerable antenna - 46 m
- Ionospheric parameters
- ~90-1000+ km
- Campaign basis
- ~1000 hours per year

Part of the MIT Haystack Observatory Designated to ionospheric and space weather research

Incoherent Scatter Radars



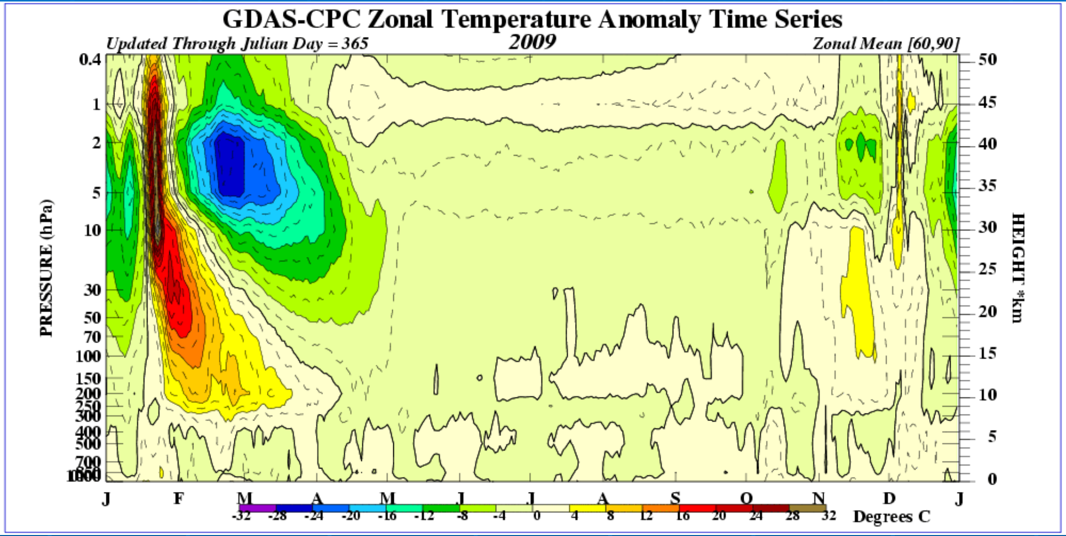
- Physical Properties of the Space Environment
  - Electron density, electron temperature, ion temperature, plasma velocity, neutral wind
  - Ne, Te, Ti, Vi – 90-700km; Un, Vn – 95-130km; Un – F-region (200-600km)

Studies of vertical coupling in the ionosphere: motivation and approach

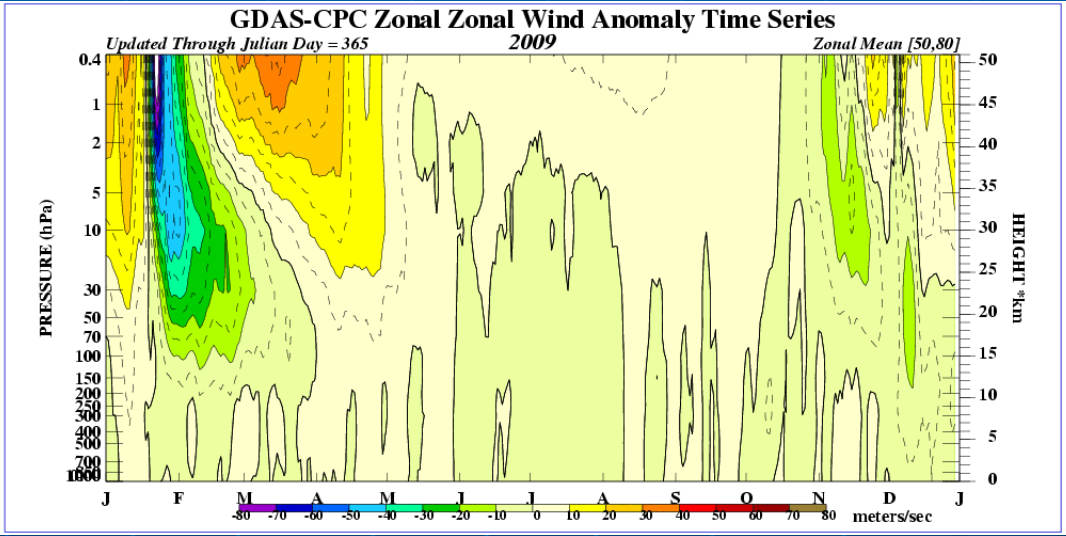
- Motivation: understanding day-to-day ionospheric and thermospheric variability
- Substantial variability during geomagnetically quiet time: ~20% at daytime, ~33% at night (*Rishbeth and Mendillo, 2001, Forbes et al., 2000, Mendillo et al., 2002*)
- Recent studies present evidence of thermospheric and/or ionospheric disturbances due to tsunami, thunderstorms, earthquakes, tides – i.e. coupling from below
- Target events: stratospheric sudden warmings

"The search for links between F-region phenomena and the lower atmosphere has a long history and few real outcomes"
H. Rishbeth, JASTP, 2006

SSW of January 2009



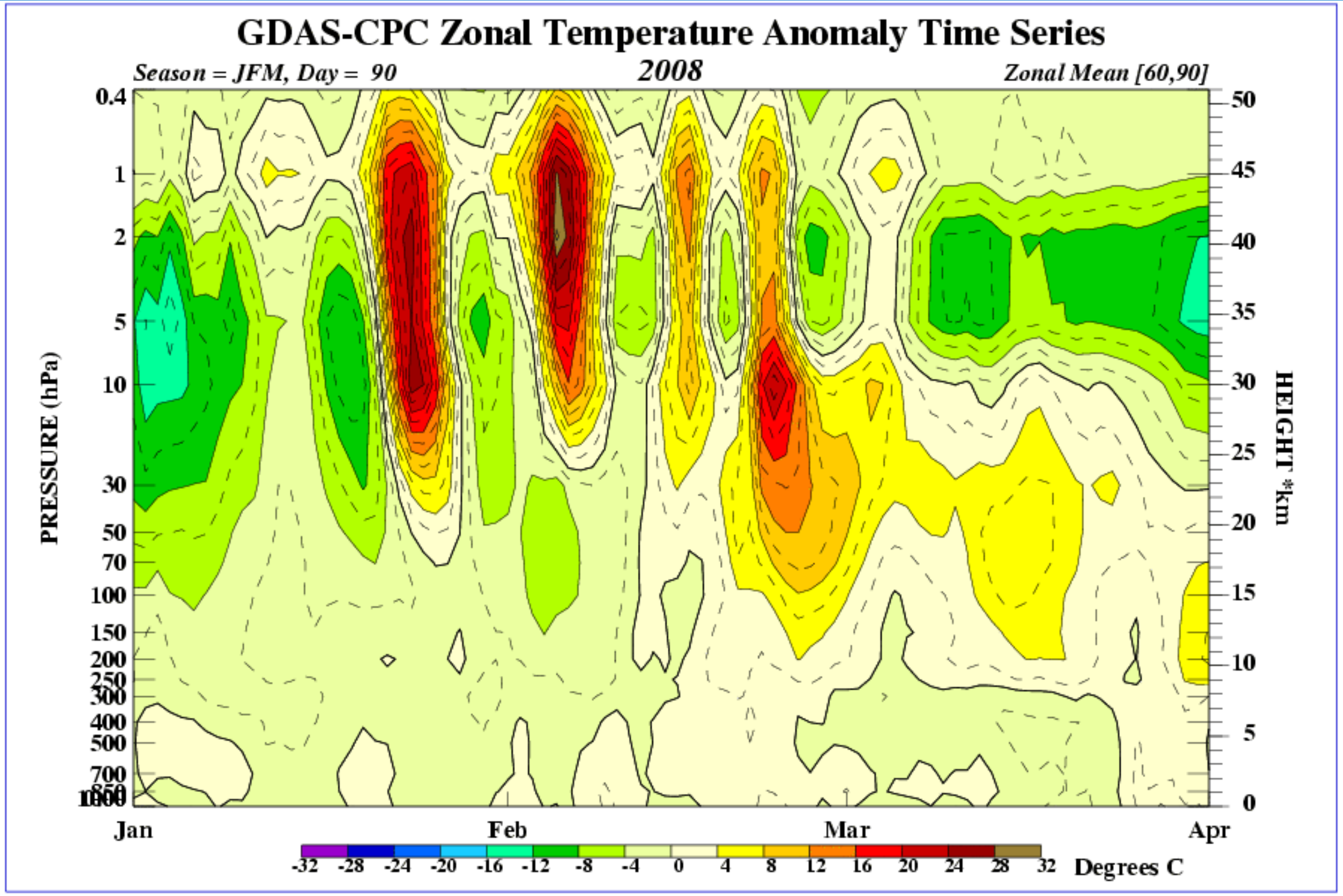
- Temperature anomaly in the polar stratosphere: increase in temperature by 40-70K
- Results from interaction of quasi-stationary planetary wave with zonal mean flow



- Wind anomaly in the polar stratosphere: zonal mean zonal wind abates or changes direction from winter-type to summer-type

SSW is a large-scale wintertime phenomenon in the polar stratosphere

Jan 2008 – Mar 2008



Some SSW propagate downward in 2-3 weeks and may effect the weather

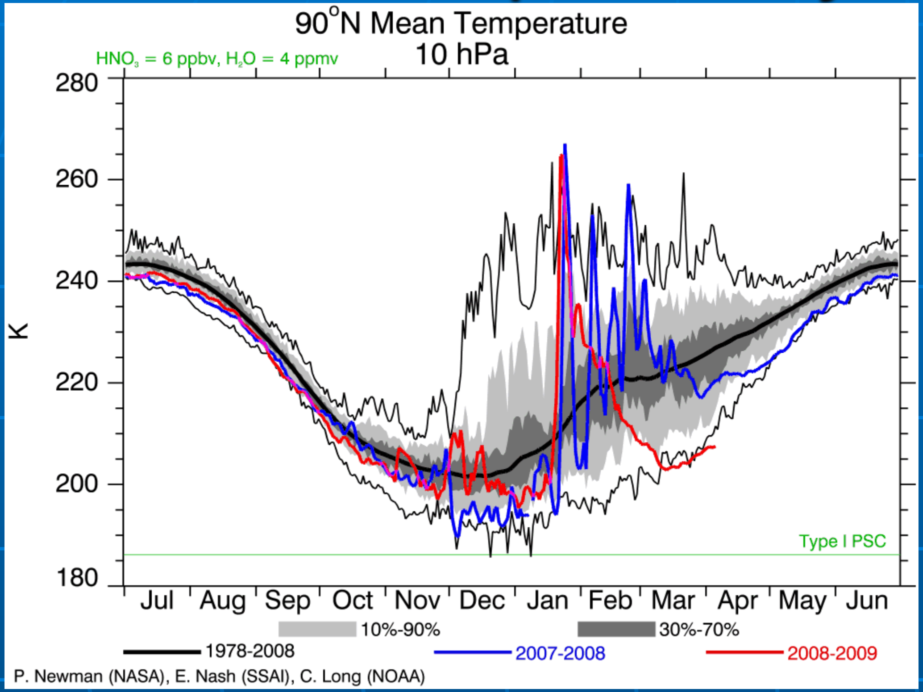
Reports of mesospheric cooling associated with stratospheric warming

Effects of SSW at higher altitudes (lower thermosphere > 100km, thermosphere/ ionosphere) were largely unknown until 2008

Why stratospheric warmings are a good target for studies?

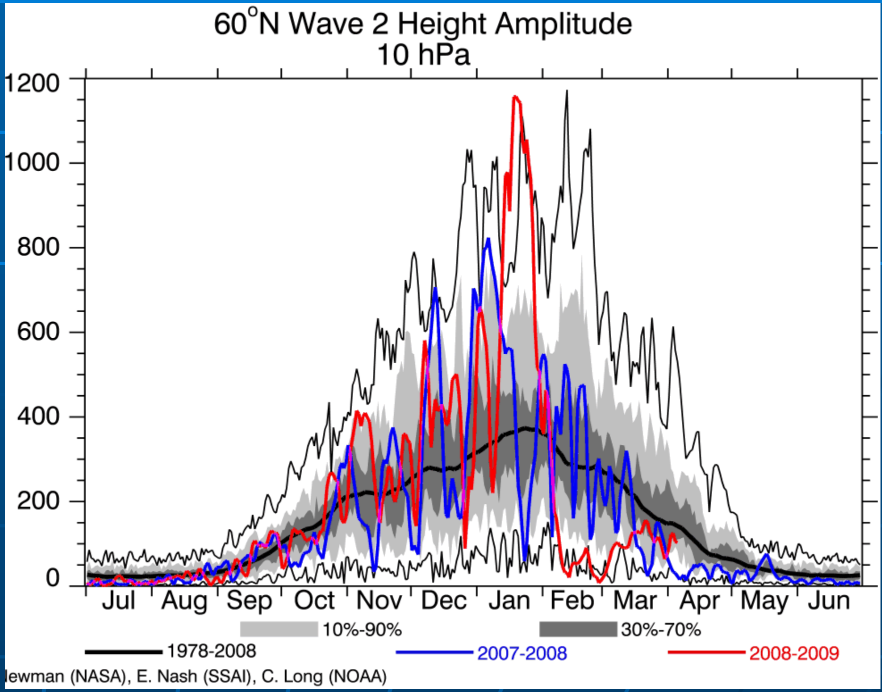
- Large-scale event – effects can be seen by many existing instruments
- Minor SSW occur several times per season – good statistics
- SSW last from few days to ~2 weeks – long enough to see effects
- 30-year continuous data on stratospheric conditions – can study already existing data
- NCEP/ECMWF forecasts – can plan future experiment

Stratospheric sudden warming and planetary wave activity



Record SSW in January 2009…

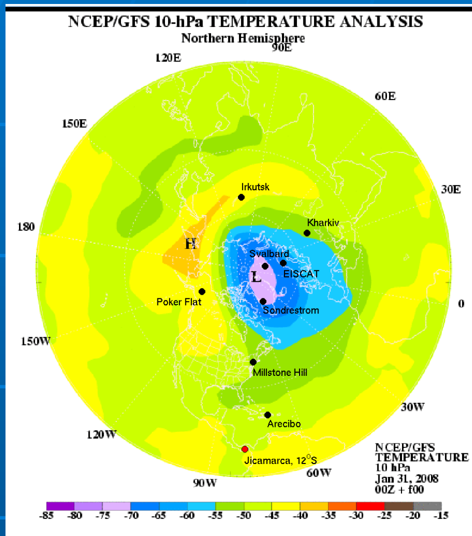
…preceded by a record increase in planetary wave 2 activity



Solar min + record stratwarming
= ideal case scenario

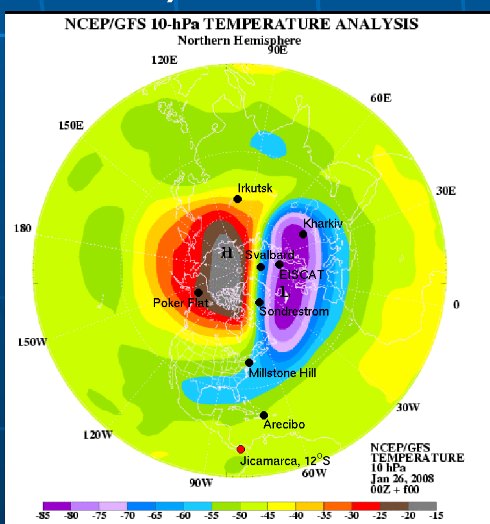
Experimental campaigns: 2008, 2009, 2010, 2011

Pre-SSW vortex

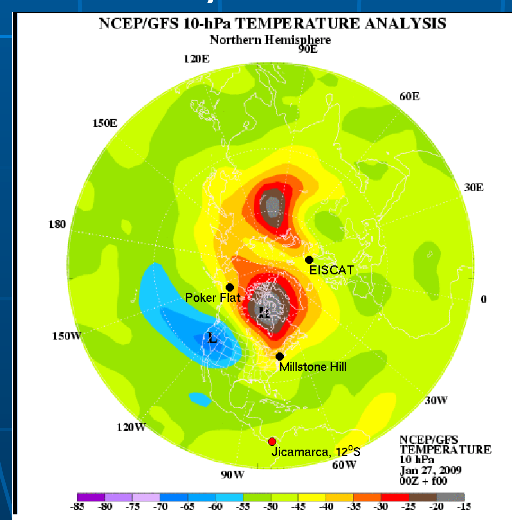


- 4 multi-day experimental ISR campaigns during SSW events: January 2008, January 2009, January 2010, February 2011
- Data coverage and operation modes vary for different radars; different radars probe ionosphere above cold or warm stratospheric cells
- Radar campaigns in the alert mode based on 8-10 days stratospheric forecasts
- 4 successful campaigns out of 4 attempts

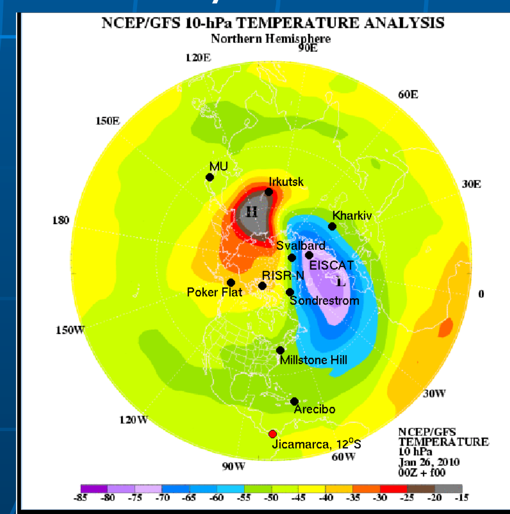
January 2008 SSW



January 2009 SSW

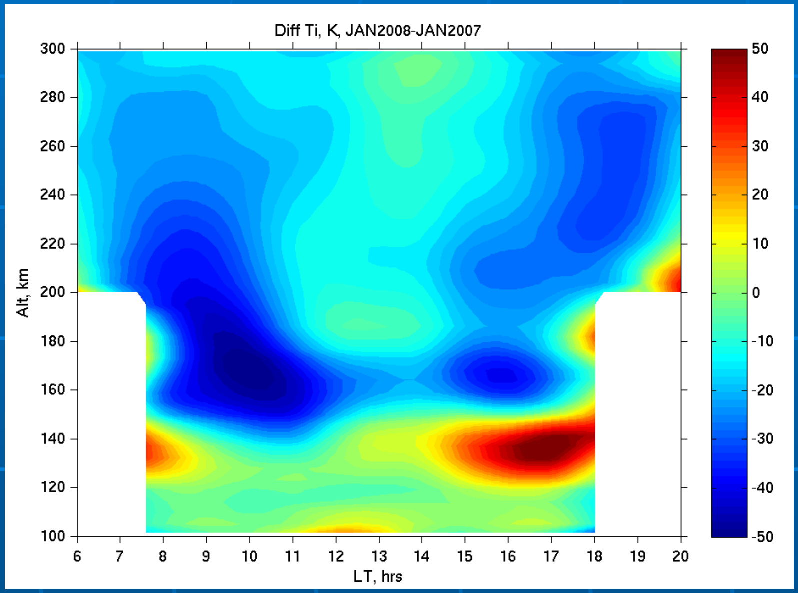


January 2010 SSW



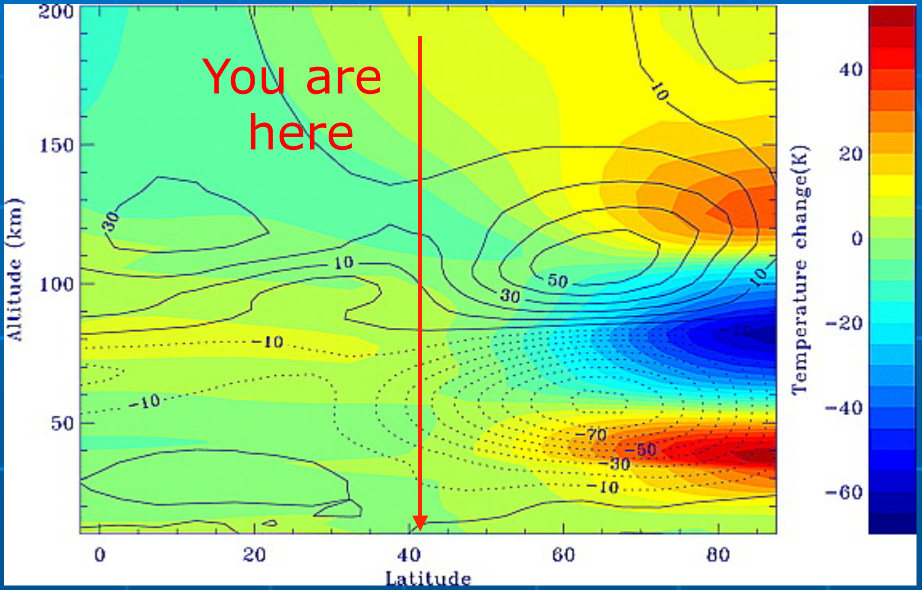
Temperature variations during SSW

Data: Millstone Hill ISR, 42°N



Goncharenko and Zhang, 2008

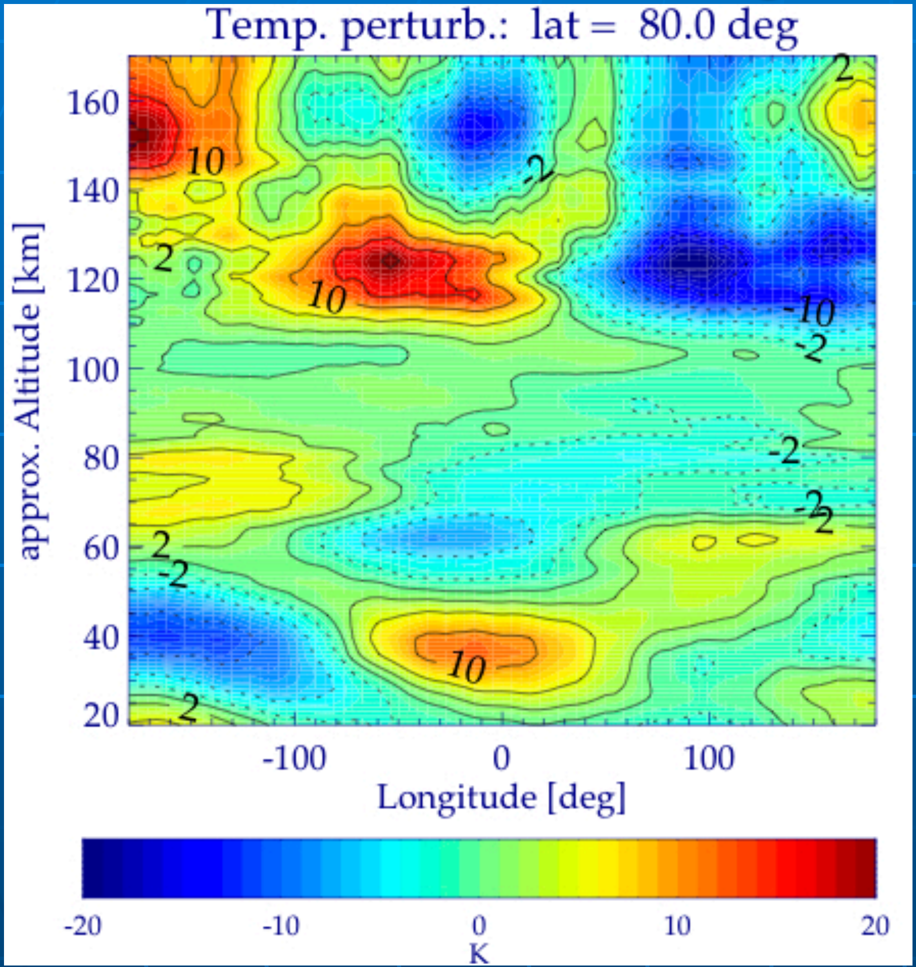
Model: TIMEGCM



Liu and Roble, 2002

- Data: warming at 120-140km; cooling above ~150 km; 12-hour wave;
- First experimental evidence of alternating warming and cooling of upper atmosphere
- Model: mesospheric cooling and secondary lower thermospheric warming

Other experimental evidence of temperature change during SSW

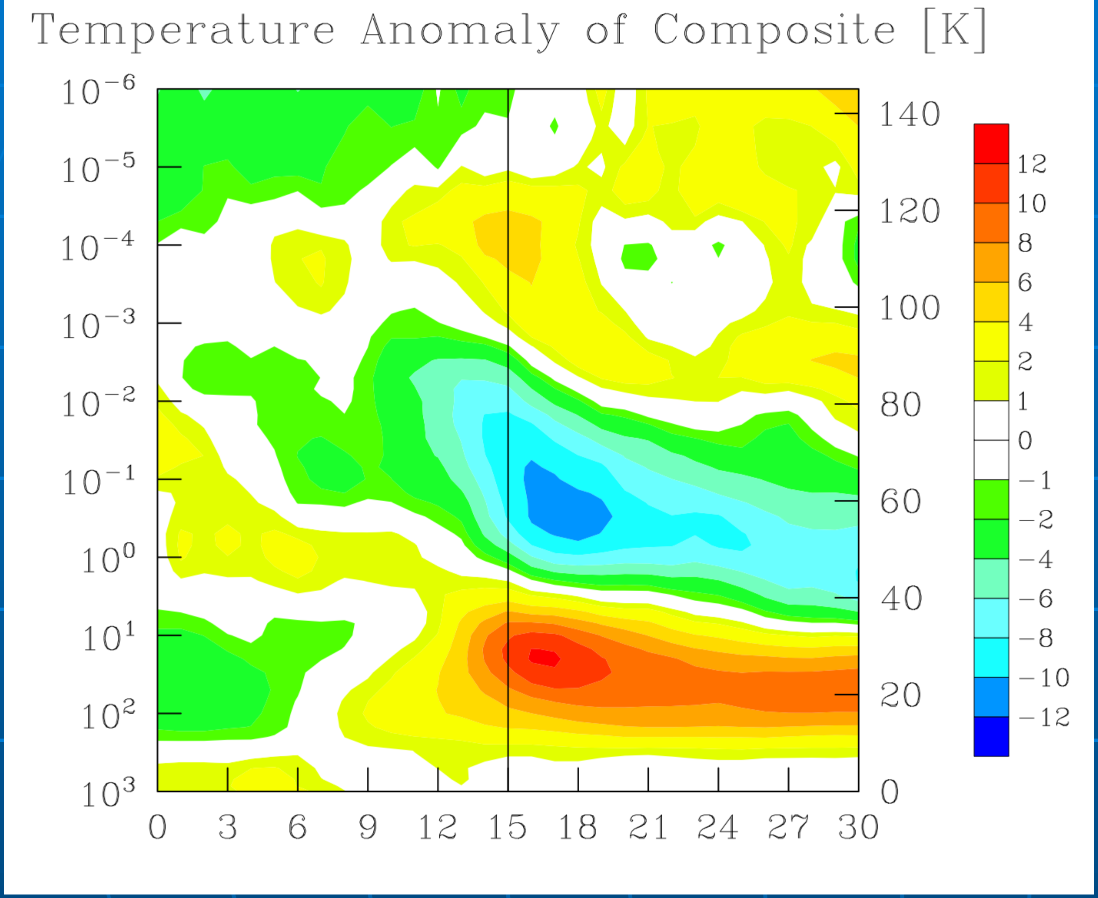


Funke et al., 2010

- MIPAS on ENVISAT – Tn increase at 120-140 km during Jan 2009 SSW
- wave 1 pattern
- Ti increase at 120-142 km in EISCAT data (*Kurihara et al, 2010*)
- Tn and Ti decrease in the F-region at Poker Flat – FPI and ISR data (*Conde and Nicolls*, 2010)

Three new studies demonstrate warming at 120-150 km or cooling in the F –region related to SSW

Modeling evidence of Tn change



Courtesy of Andy Miller and Hocke Schmidt

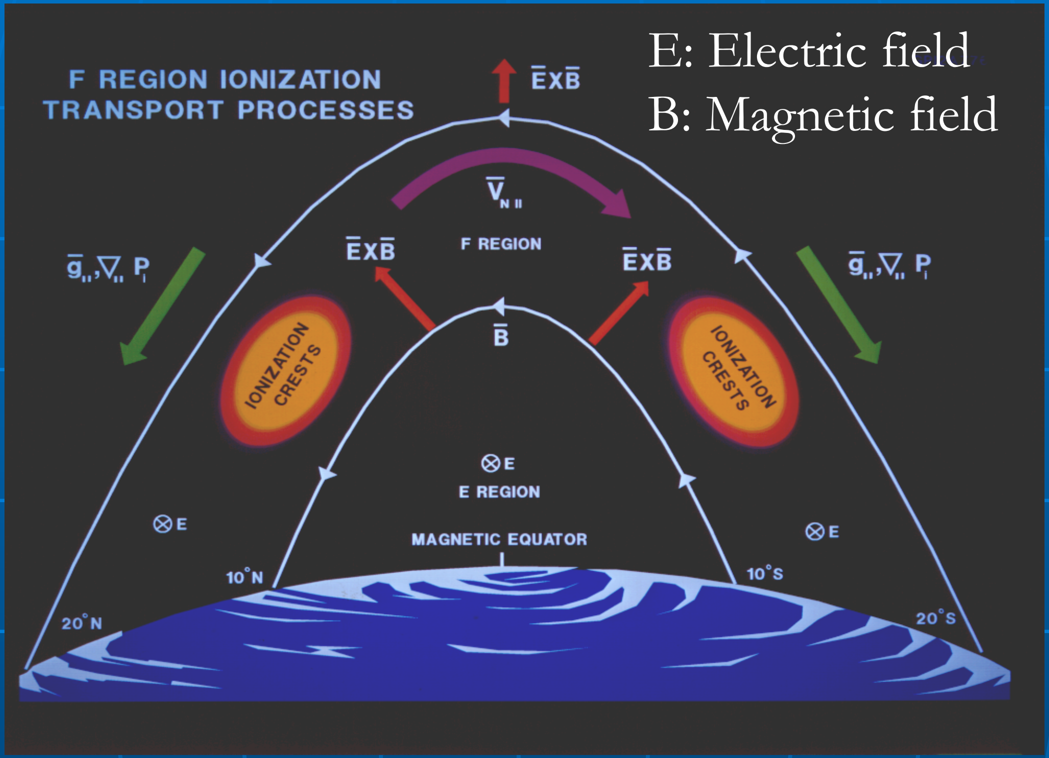
- MPI- HAMMONIA model
- Composite run, 20 major SSW events
- Multi-day warming in the MLT region

Other simulations:

- TIMEGCM, *Liu et al., 2010*
- WAM, *Fuller-Rowell et al., 2010*
- Kühlungsborn Mechanistic Climate Model, *courtesy of Christoph Zülicke*
- TIMEGCM, *Yamashita et al., 2010* - role of gravity wave spectrum for lower thermosphere Tn change

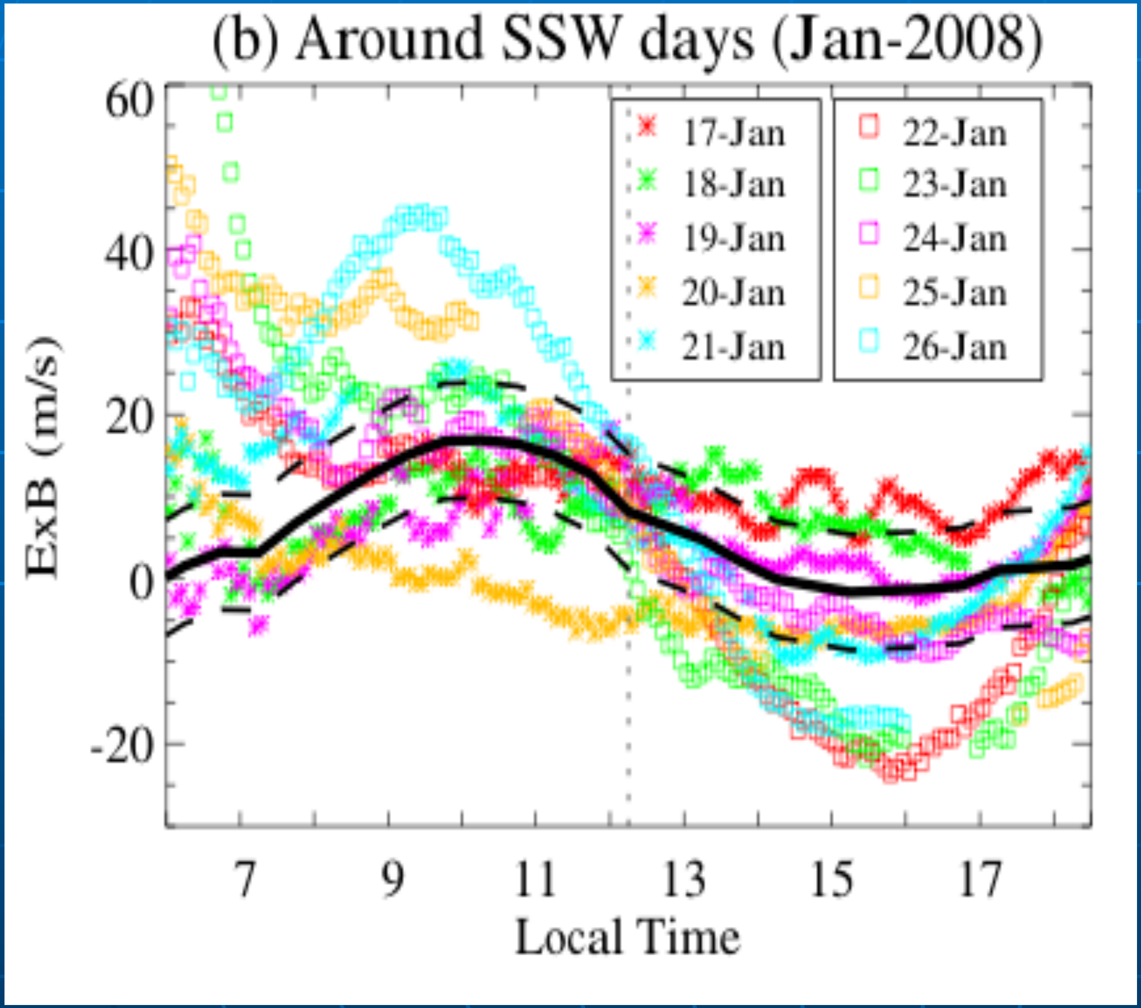
Four different models predict lower thermospheric warming during SSW

Low Latitude Ionosphere



- Highly sensitive to vertical transport
- The driver of vertical motion remains an unresolved problem

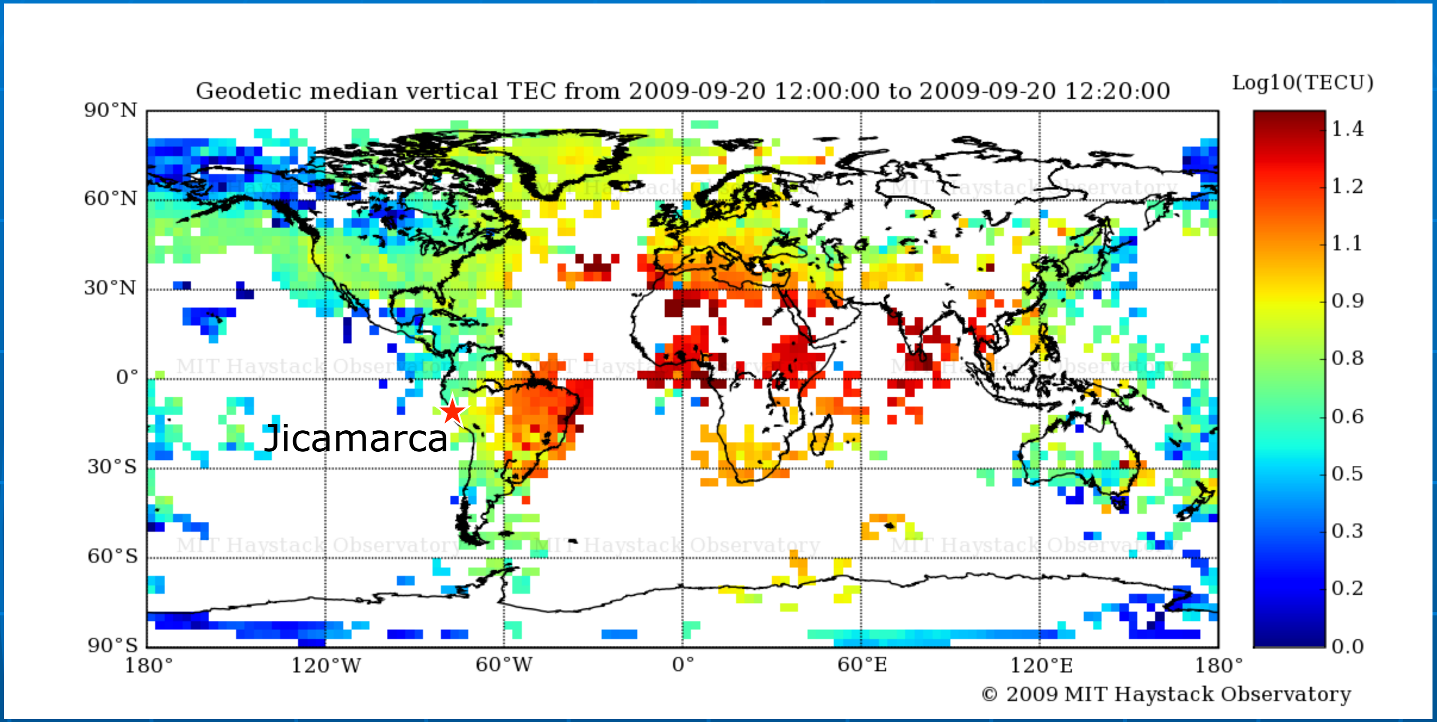
Vertical drift over Jicamarca, Jan 2008 SSW event



- Large enhancements in the vertical drift
- Upward drift in the morning, downward in the afternoon
- Peaks few days after the peak in stratospheric temperature
- Persistent for several days
- Similar effects in drifts for other SSW events

Chau et al., 2009

GPS TEC data used



1. GPS TEC, MIT Haystack Observatory:
   - ~2000 GPS receivers,
   - 5 min, 1ºx1º resolution
   - Focus on low latitudes in the Western hemisphere; use Jicamarca radar (12ºS,75ºW, magnetic equator) vertical drifts
   - LISN TEC data enable extension of this work in the Southern America

January 2009 SSW: Jicamarca ISR and GPS TEC

15 UT mean

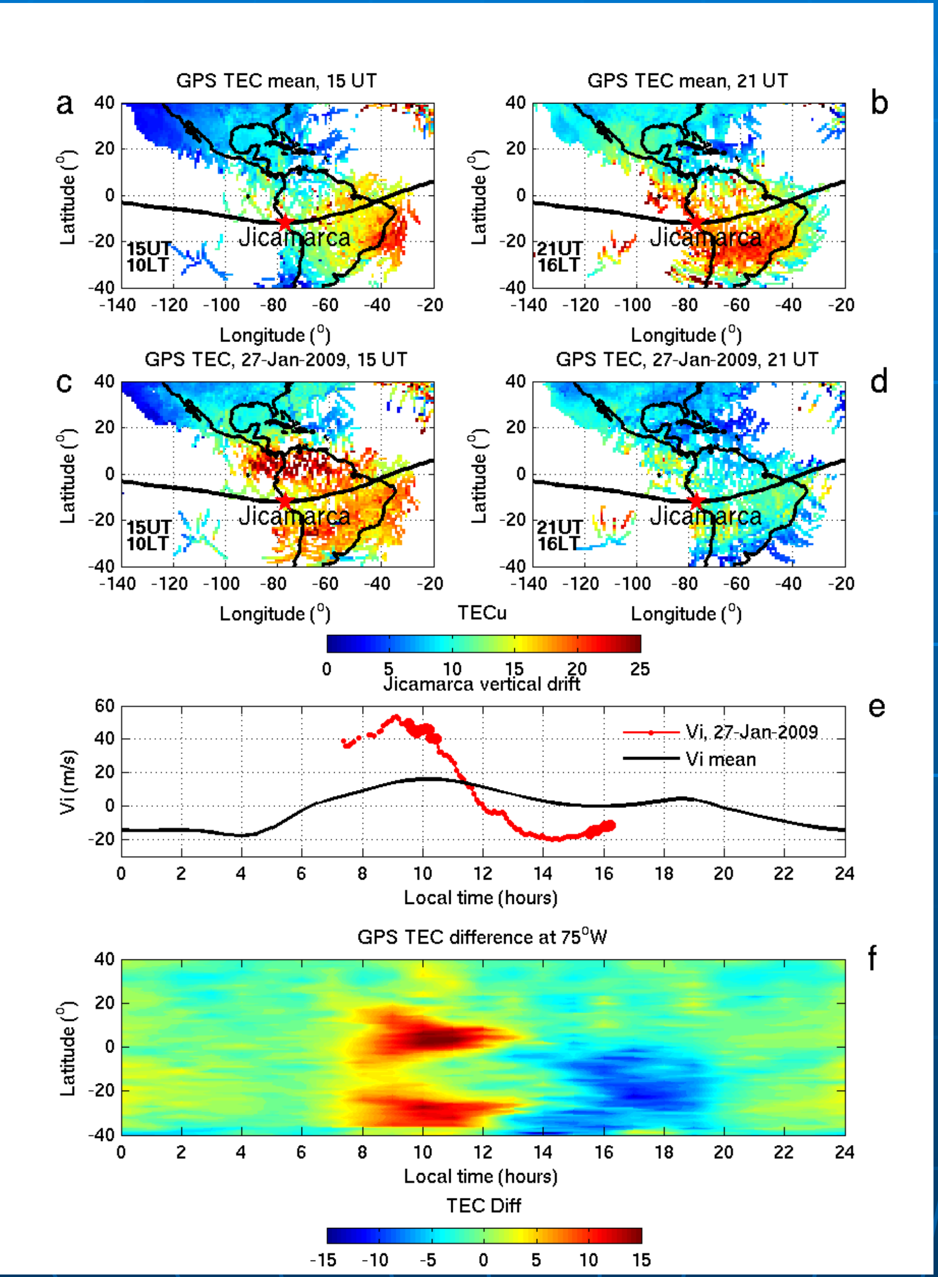
21 UT mean

15 UT SSW

21 UT SSW

Jicamarca drift

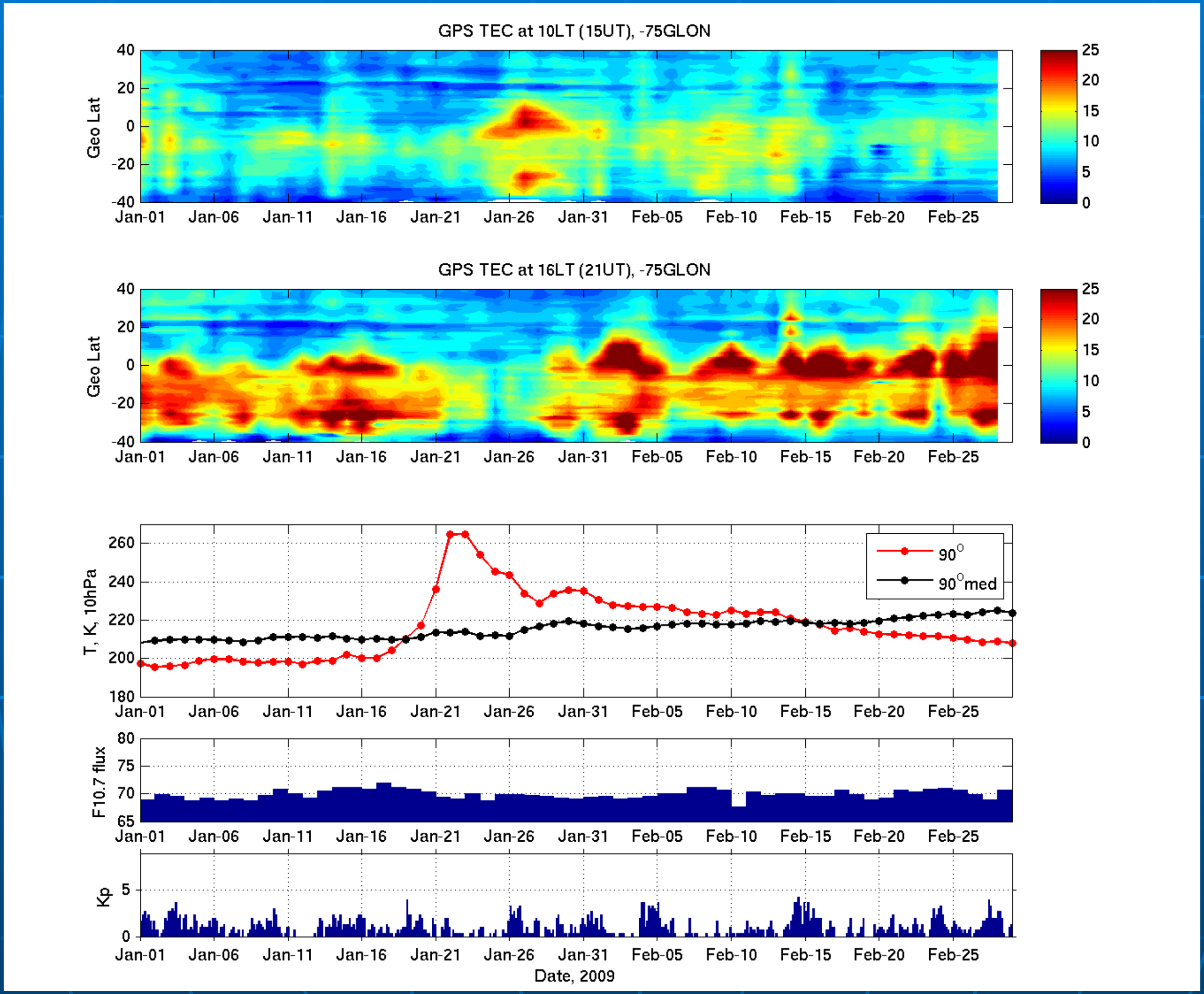
TEC change



- Entire daytime low to mid-latitude ionosphere is affected during stratwarming
- 50-150% variations in TEC

Goncharenko et al., 2010a

GPS TEC data at 75°W for the winter of 2009



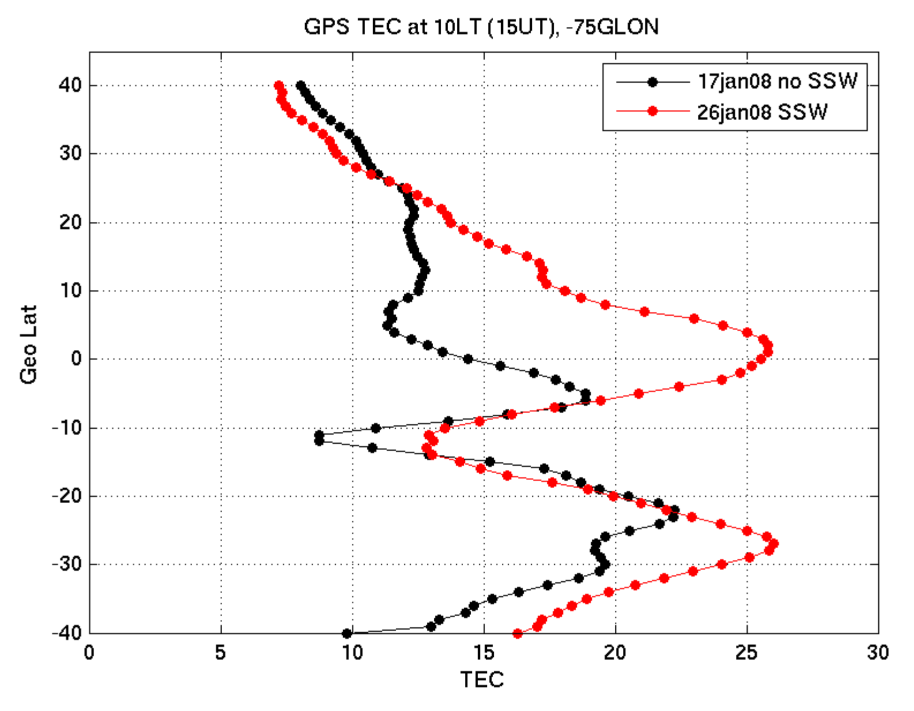
10 LT

16 LT

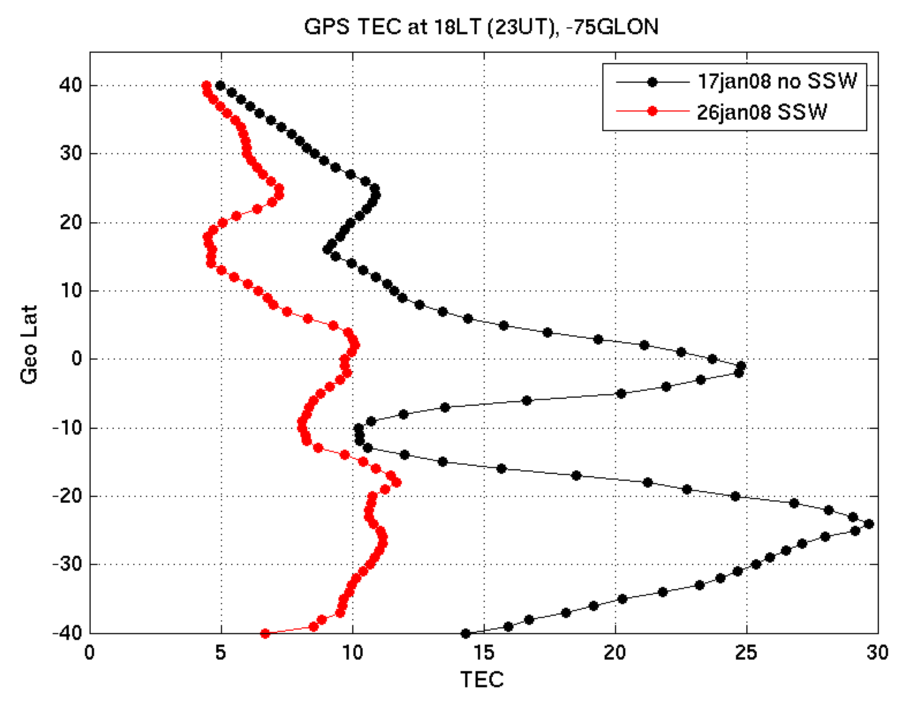
Lower atmospheric forcing during SSW is the only conceivable reason for the observed ionosperic variations.

GPS TEC at 75ºW

Morning sector



Afternoon sector

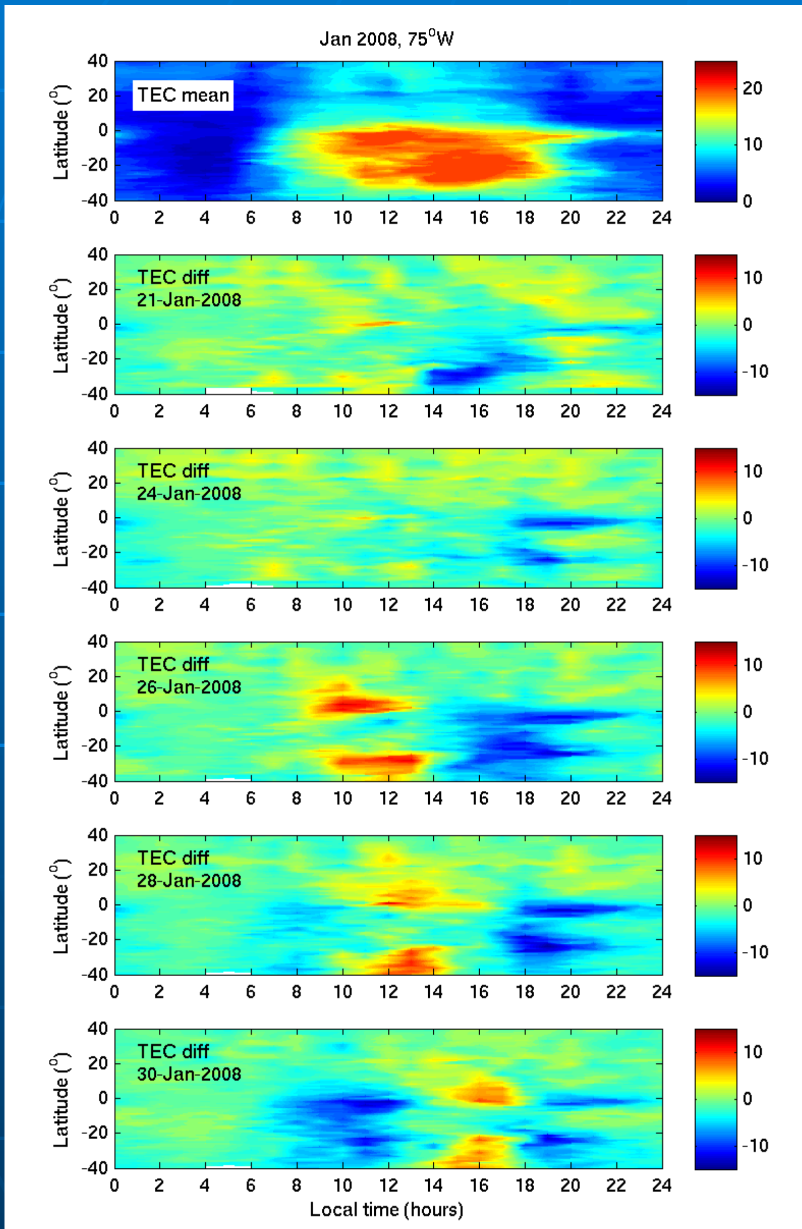


- Equatorial anomaly is enhanced in the morning sector by 20-40%
- Crest locations moved to higher latitude

Equatorial anomaly is suppressed in the afternoon sector by a factor of 2-3

During the stratospheric warming, equatorial fountain is turned on in the morning and turned off in the afternoon

GPS TEC at 75^{o}W, Jan 2008



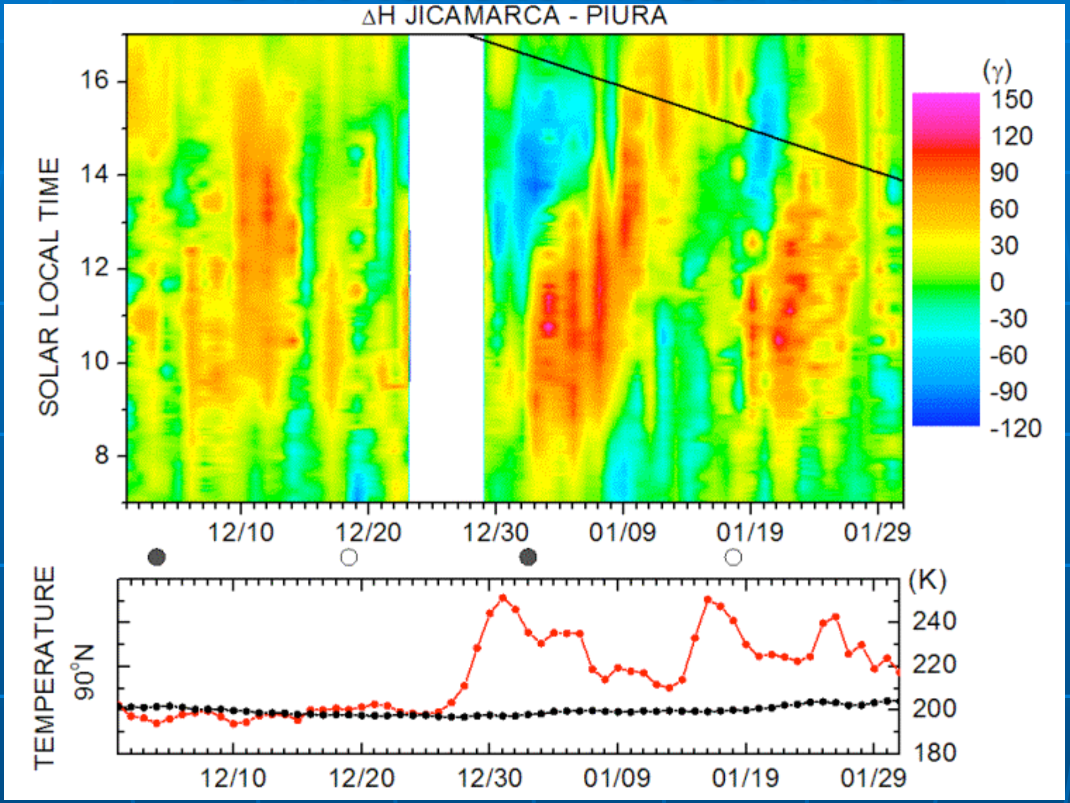
No SSW

SSW

- Peak daytime TEC: 15 TECu
- No clear pattern prior to SSW
- Variation in TEC during stratwarming: semidiurnal wave, ~5-12 TEC
- Progressive shift to later local times
- Both high amplitude of the wave and rapid phase change lead to large variability in TEC

Goncharenko et al., 2010b

Temporal development of ionospheric response to SSW: shift to later local time

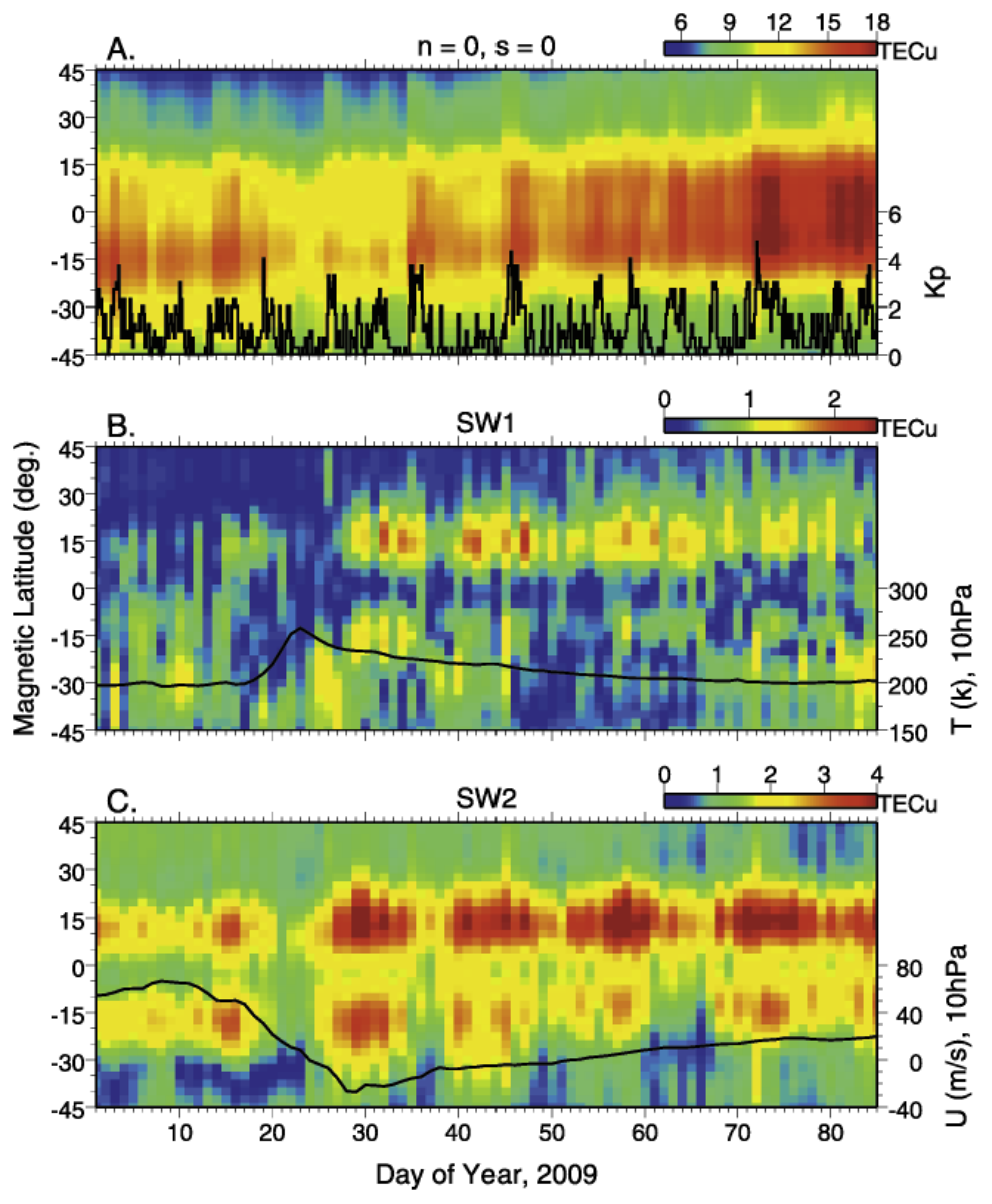


Fejer et al., 2010

- 12-h wave response progressively shifts to later local times
- Reported in CHAMP and ground-based magnetometers (*Fejer et al*., 2010), multiple SSW events

Both large magnitude of 12-h wave and a rapid phase change are responsible for the dramatic variability of the ionosphere

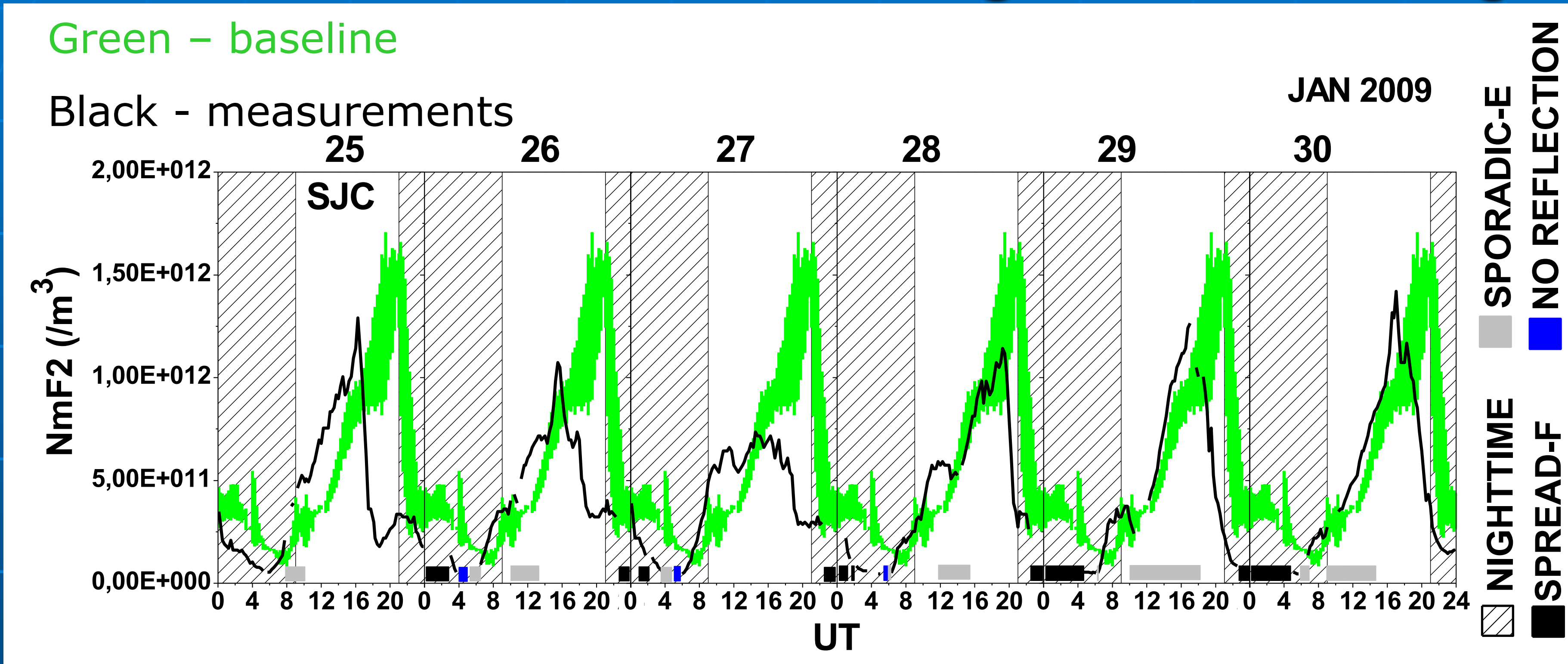
Tidal variations in TEC during Jan 2009 SSW



- JPL TEC dataset (2.5°x2.5°, 2 hour resolution, global dataset)
- Increase in migrating and non-migrating 12-h tide during SSW

Pedatella and Forbes, 2010

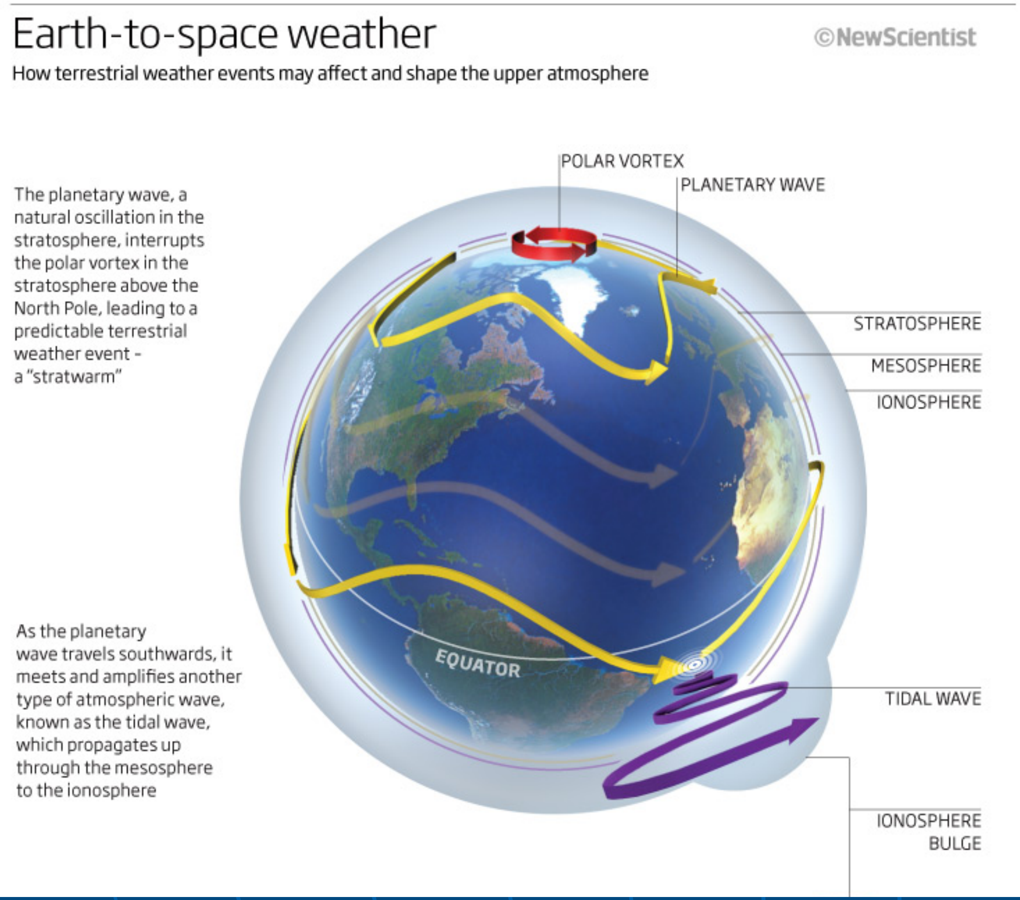
South America: Variations in NmF2 during stratwarming



Courtesy of Y. Sahai and R. de Jesus

- São José dos Campos digisonde (23.2 S, 45.9 W)
- January 2009 SSW event
- Decrease in NmF2 by a factor of 4 at sunset; persistent for several days

Mechanism: interaction of planetary wave with tide

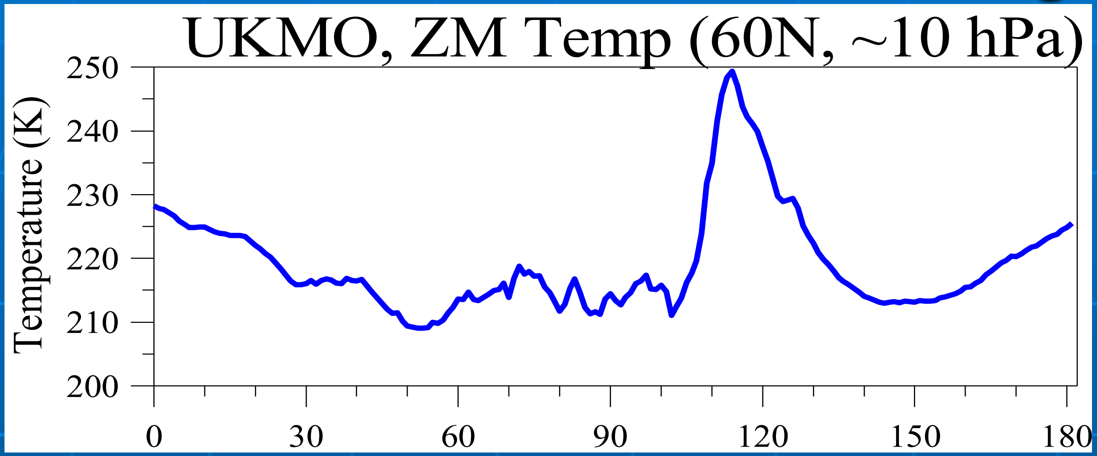


New Scientist article: Phantom storms: How our weather leaks into space
06 October 2009 by
Jon Cartwright

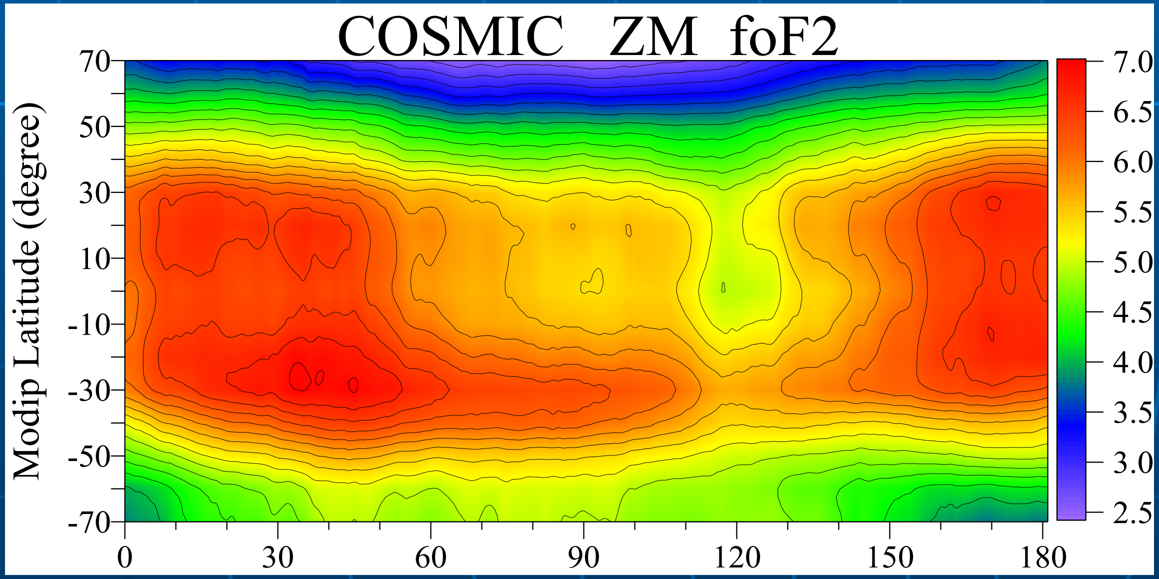
Non-linear interaction of planetary wave with semidiurnal tide results in:
- Modulation of the 12-h wave amplitude and E-region dynamo
- Generation of non-migrating 12-h tide

Main difficulty: propagation of quasi-stationary planetary waves is restricted in altitude and latitude

Zonal mean variations during stratospheric warmings

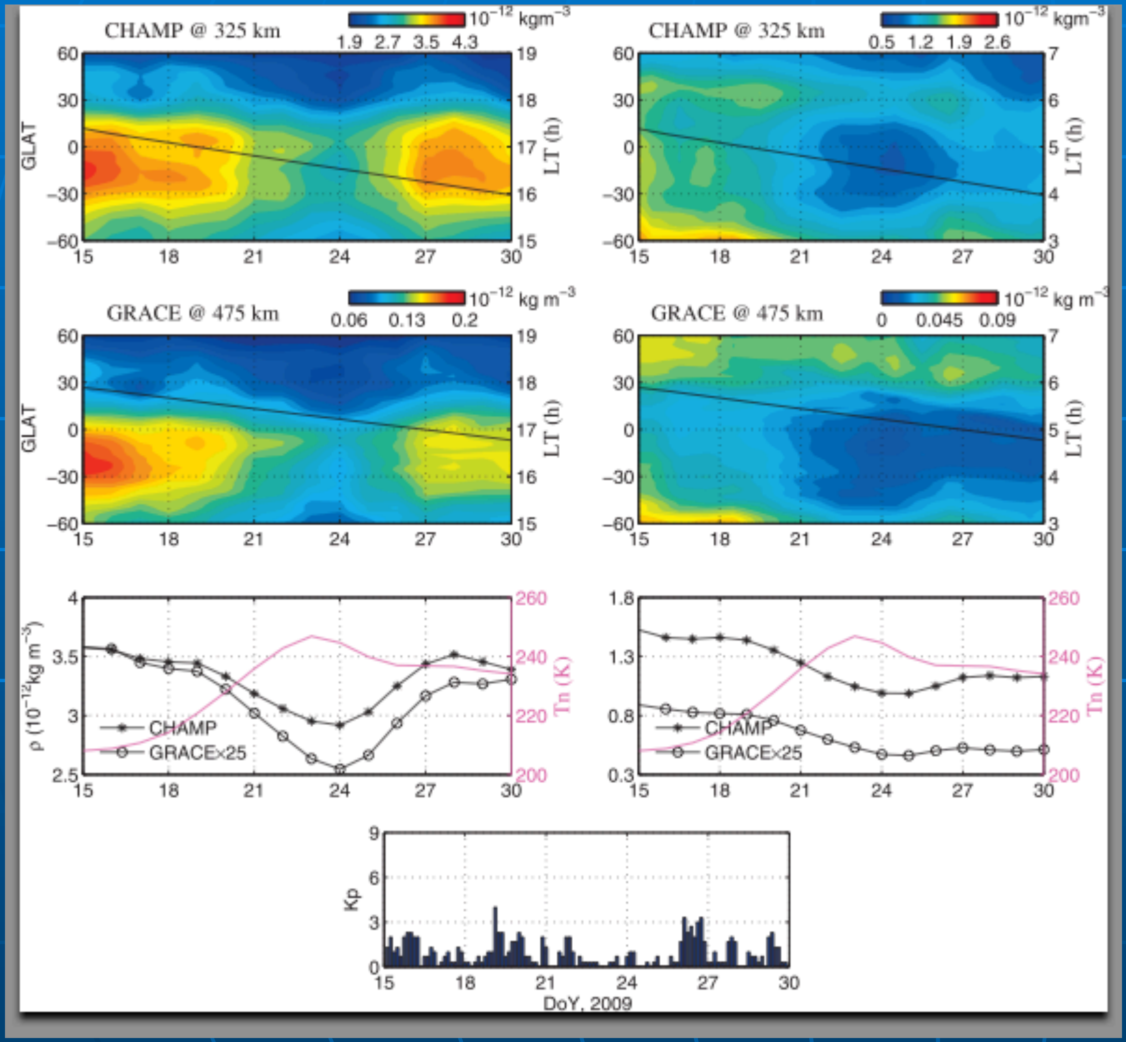


- •COSMIC foF2 and TEC data
- •Decrease in zonal mean foF2 and hmF2; decrease in 24-h component
- •Multiple SSW events



Pancheva and Muhtarov, 2011

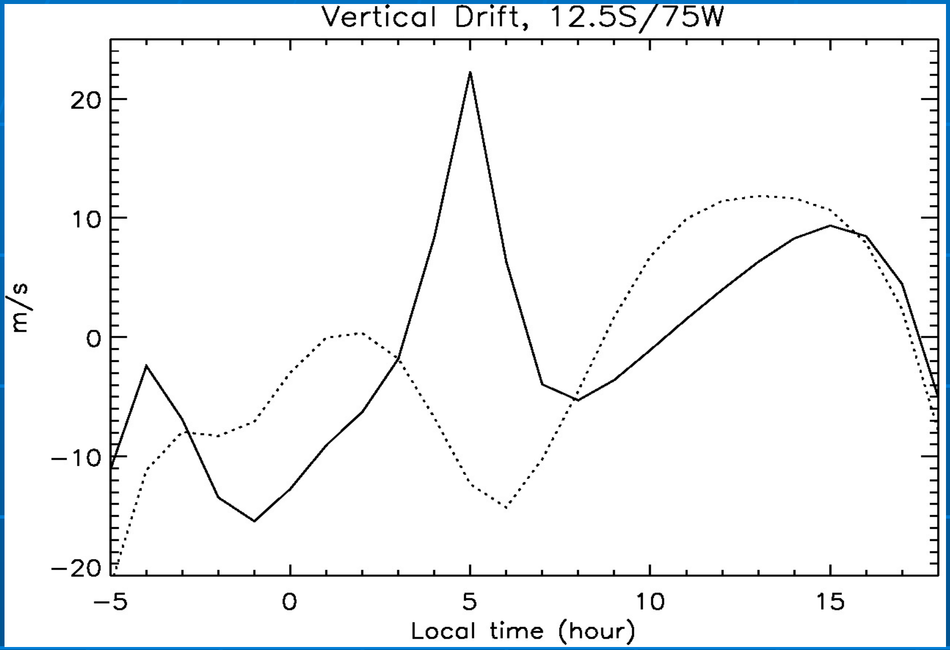
Thermospheric density during SSW Jan 2009

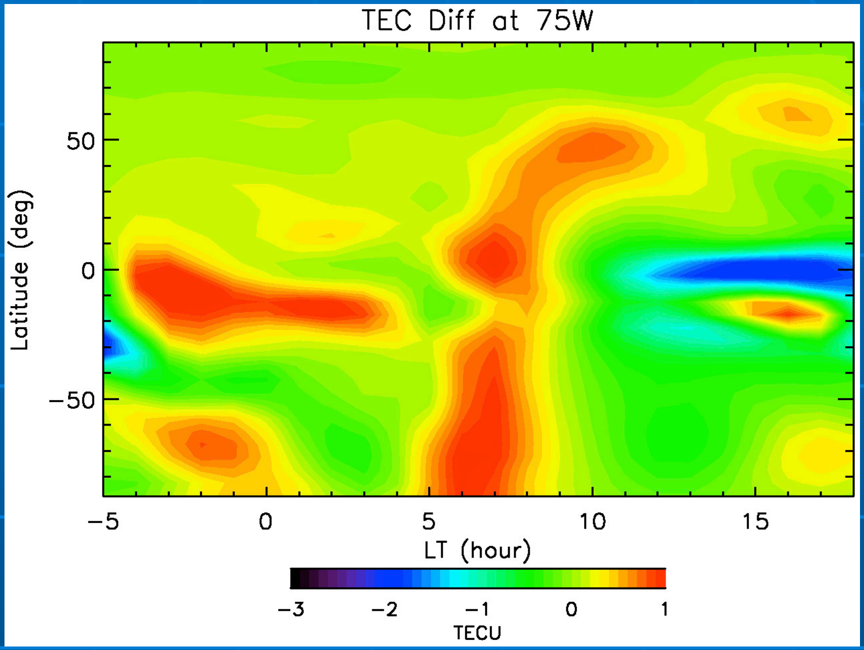


Liu et al., 2011

- CHAMP, 325 km: decrease by ~30%
- GRACE, 475 km: decrease by ~45%
- Accompanied by cooling ~50K
- Significant longitudinal differences

TIMEGCM simulations: Effects of Planetary Waves in the Ionosphere

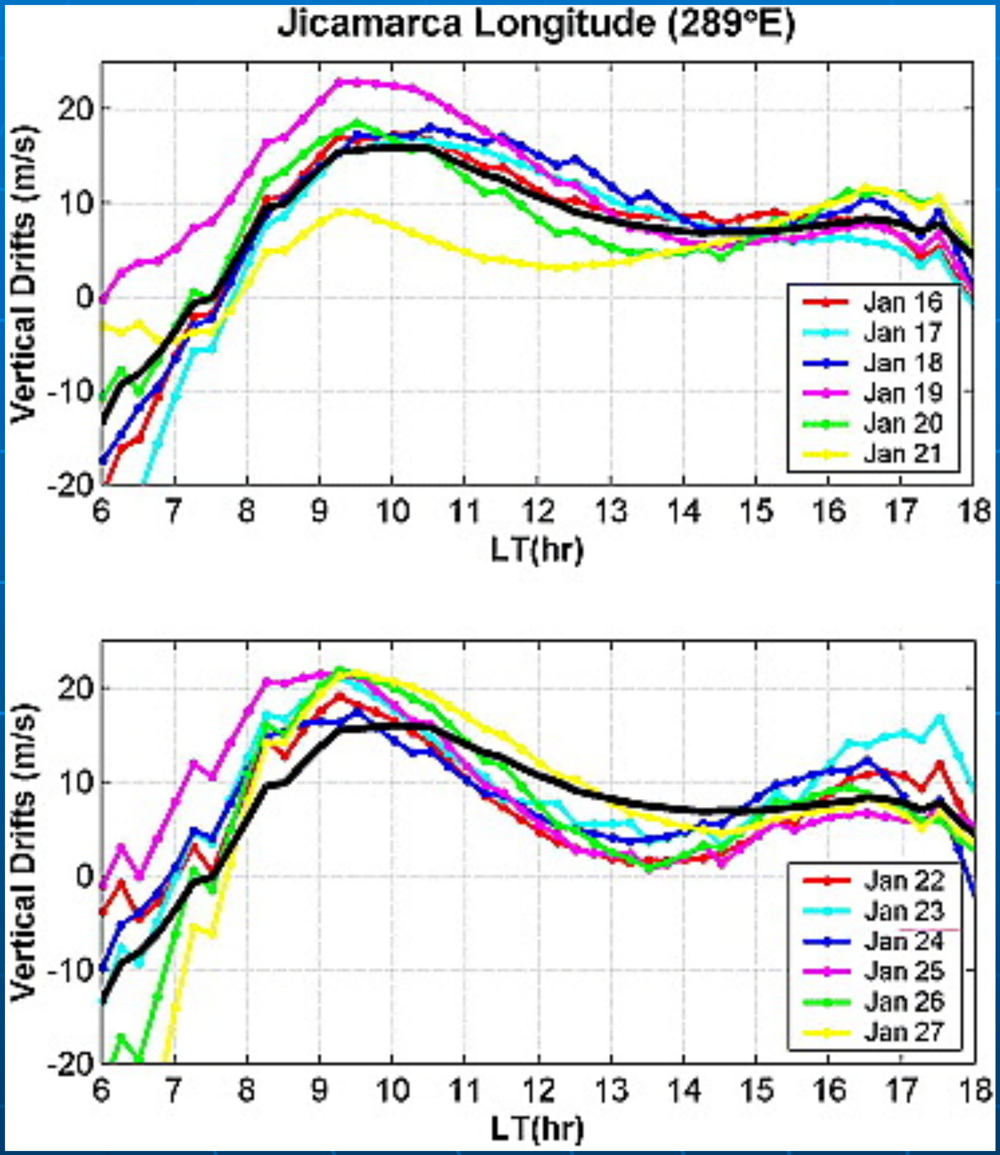




- Planetary wave introduced at 60°N; solar min conditions
- Large electric fields change around sunrise
- Total electron content change
- Consistent with observations by Jicamarca ISR and GPS TEC, but smaller magnitude

Liu et al., 2010

WAM simulation of SSW



- Cooling in the mesosphere and a warming in the lower thermosphere, consistent with observations.

- Changes in equatorial electrodynamics during daytime hours, consistent with observations

Fuller-Rowell et al., 2010

Questions arising from early results

1. What is the nature of the tide (solar or lunar?)

   •Evidence for lunar tide presented by *Fejer et al, 2010*

   •Consistent with analysis of lunar tides in GPS TEC data by *Pedatella and Forbes, 2010b*

2. What drives the temporal development of ionospheric response to SSW?

   •Ionospheric response to SSW peaks several days after the peak in high-latitude stratospheric temperature

   •Mesospheric parameters change up to two weeks prior to SSW (*Azeem, 2005*, *Hoffmann, 2002, 2007*)

3. What controls the efficiency of the stratosphere-ionosphere coupling?

   •Amplitude of quasi-stationary planetary wave? duration of the increase in the planetary wave activity? changes in the high-latitude dynamics? Changes in the low-latitude stratospheric temperature and dynamics? What are the roles of gravity waves?

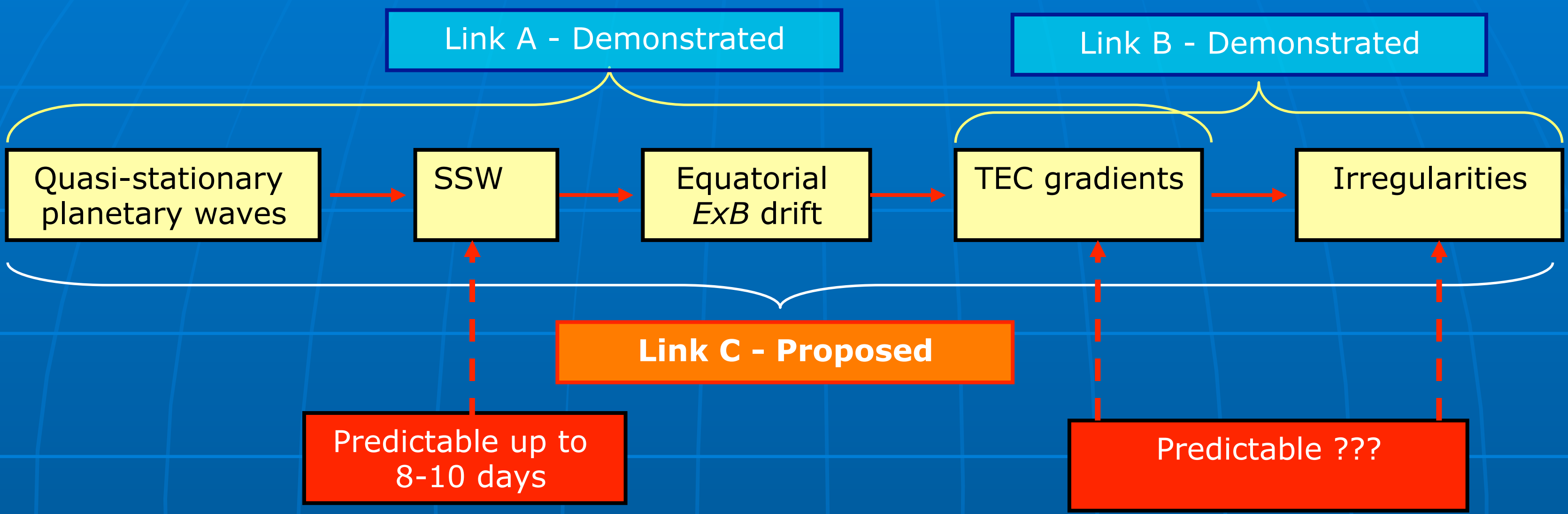
4. What kinds of tides are amplified during SSW events?

   •Migrating and/or non-migrating; 12-h, 24-h, 8-h – mixed evidence

Question: If ionospheric response is so strong, how come scientists did not see it earlier?

- **Answer 1**: Better data availability and better global models
- **Answer 2**: Solar minimum conditions + record strong stratwarmings enabled identification of SSW effects
- **Answer 3**: They did. Nobody believed them:
  - Manson et al., ~1970s; Kazimirovskiy et al., 1970-1980s, Stening, 1990s
- **Answer 4**: "Gorilla on a basketball court effect":
  - Inattentional blindness phenomenon
  - ~50% of busy audience does not see obvious objects
  - Our prior beliefs, interests and expectations shape the way we perceive the world

Possible implications

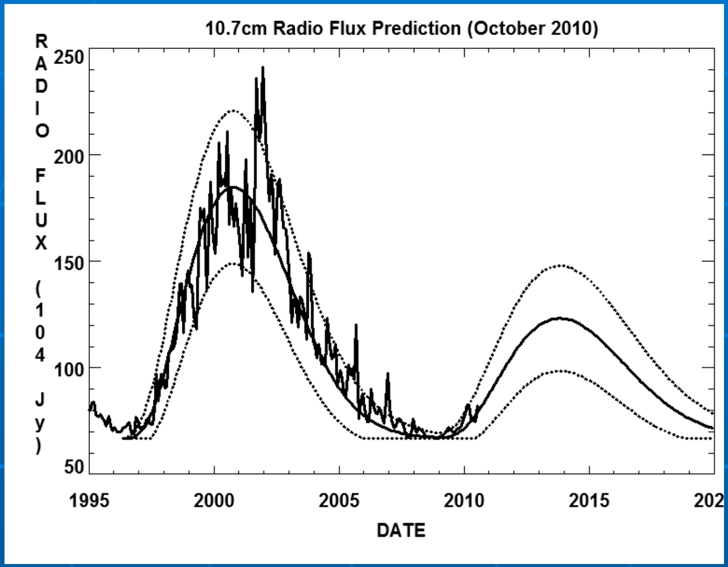


- Planetary waves can modify background ionosphere and provide pre-conditioning important for variety of mechanisms affecting irregularities
- Multiple evidence that PW-tide interaction leads to increased tidal variability in the MLT region [*Hagan and Roble*, 2001, *Liu and Roble,* 2002, *Lieberman et al*., 2004, *She et al*., 2004, *Liu et al*., 2007, *Palo et al*., 2007, *Forbes et al*., 2008, *Chang et al.*, 2009]; implications for F-region are not understood
- SSW is a “proof of concept”; applicable to other planetary waves
- Path to the multi-day ionospheric forecast

Focused studies of SSW have the potential of bringing transformative change to ionospheric research and address critical societal need

Outlook for the future

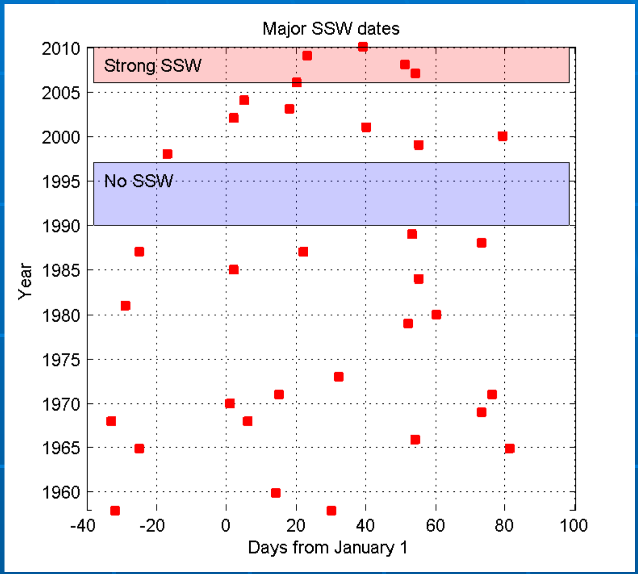
1. Solar cycle



- Low solar flux
- Easier to see phenomena coming from below

GOOD

2. SSW occurrence



- No SSW in 1990-1998; reason unknown
- Strong prolonged SSWs in 2006-2010; reason unknown

GOOD/UNCERTAIN

- Unique circumstances for the last few years; it is unknown how long they will last

Need interdisciplinary data+modeling effort with systems approach

Time to act is now

Main features of upper atmospheric response to SSW

- Warming/cooling pattern in Tn, Ti, Te
- Decrease in thermospheric density (Liu et al., 2011)
- Large variations in vertical drift at the magnetic equator (*Chau et al.*, 2009, *Anderson and Araujo-Pradere*, 2010, *Goncharenko et al*., 2010a,b, *Fejer et al*., 2010)
- Large response in Ne or TEC:
  - GPS TEC - *Goncharenko et al*., 2010
  - COSMIC Ne and TEC - *Yue et al*., 2010, *Pancheva and Mukhtarov*, 2011
- Tidal character of ionospheric response
- Persistence of response for several days
- Several days **delay** with regards to the peak stratospheric temperature
- Interaction of planetary wave with migrating semidiurnal tide generates non-migrating 12-h tide (*Pedatella and Forbes*, 2010) and amplifies tidal components (24-h, 12-h, 8-h), *Pogoreltsev*, 2007, *Liu et al*., 2010, *Fuller-Rowell et al*., 2010

Summary

- Clear and rapidly growing observational evidence of lower atmosphere connection with the upper atmosphere during SSW
- **Consistency** between different types of observational techniques and analysis methods
- **Repeatability** of main features for different SSW events
- Qualitative agreement between simulations and experimental observations.
- Strongest ionospheric effects are observed at low latitudes; 50-150% change in TEC; factor of 4 change in NmF2
- Major repeatable features: 12-h tide, persistence, strength, few days delay from the peak of stratospheric temperature, mostly daytime hours
- Part of the emerging consensus considering lower atmospheric forcing as one of the major causes of ionospheric variability

Multi-day stratospheric forecast (8-10 days)

\+ delay in ionospheric response (3-6 days)

= path to multi-day ionospheric forecast